

CSC1042 Computer Network II



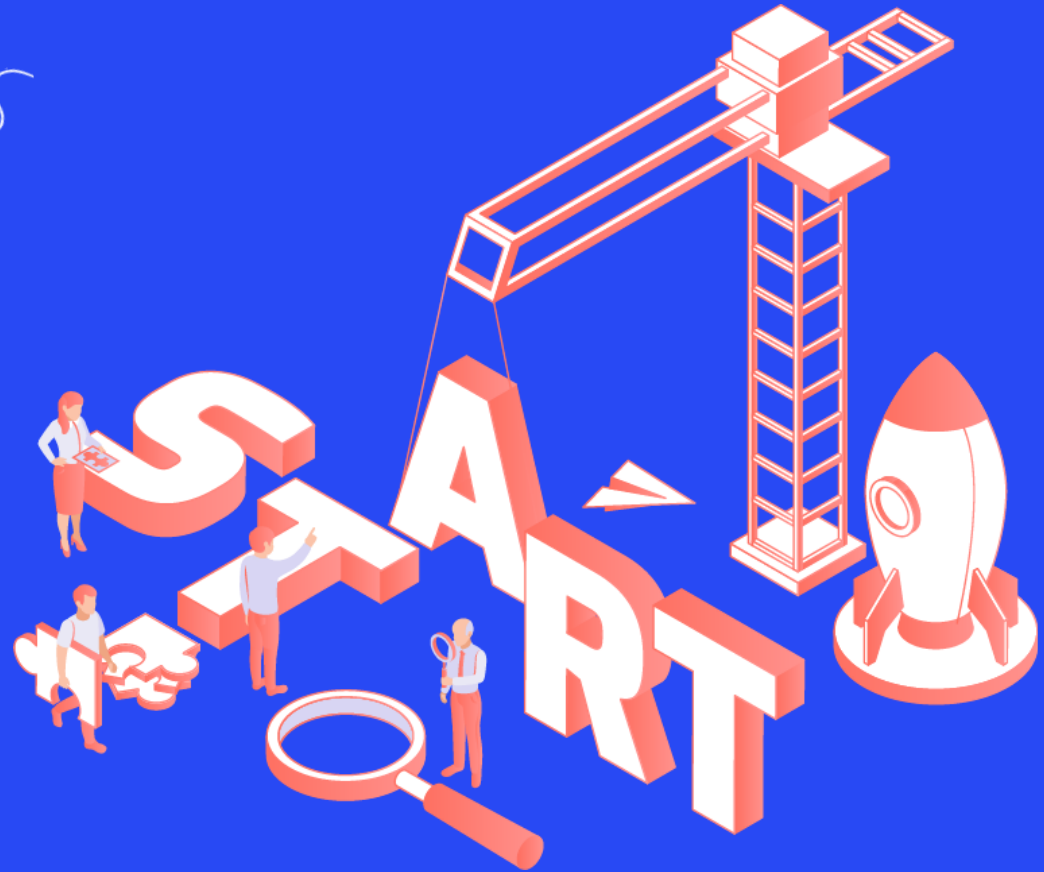
IPv4 Subnetting

Dr. Mohammed
Amine TOGOU



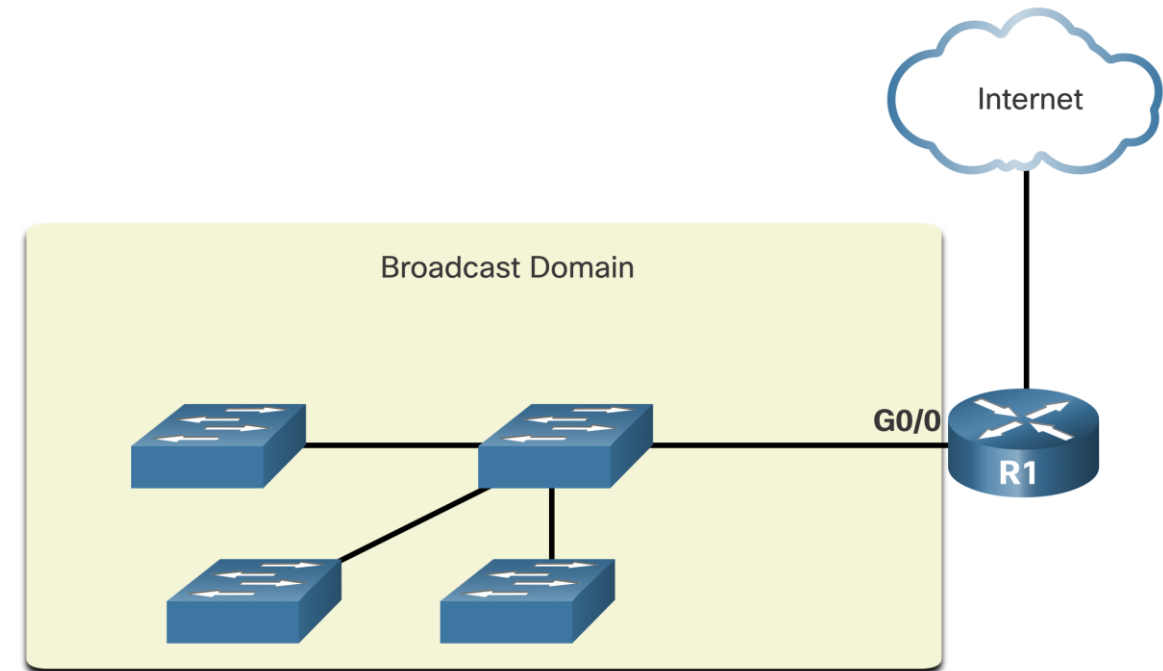
Learning outcomes

- Explain why subnetting is important
- Calculate IPv4 subnets for classes A, B, and C
- Create flexible addressing scheme using VLSM



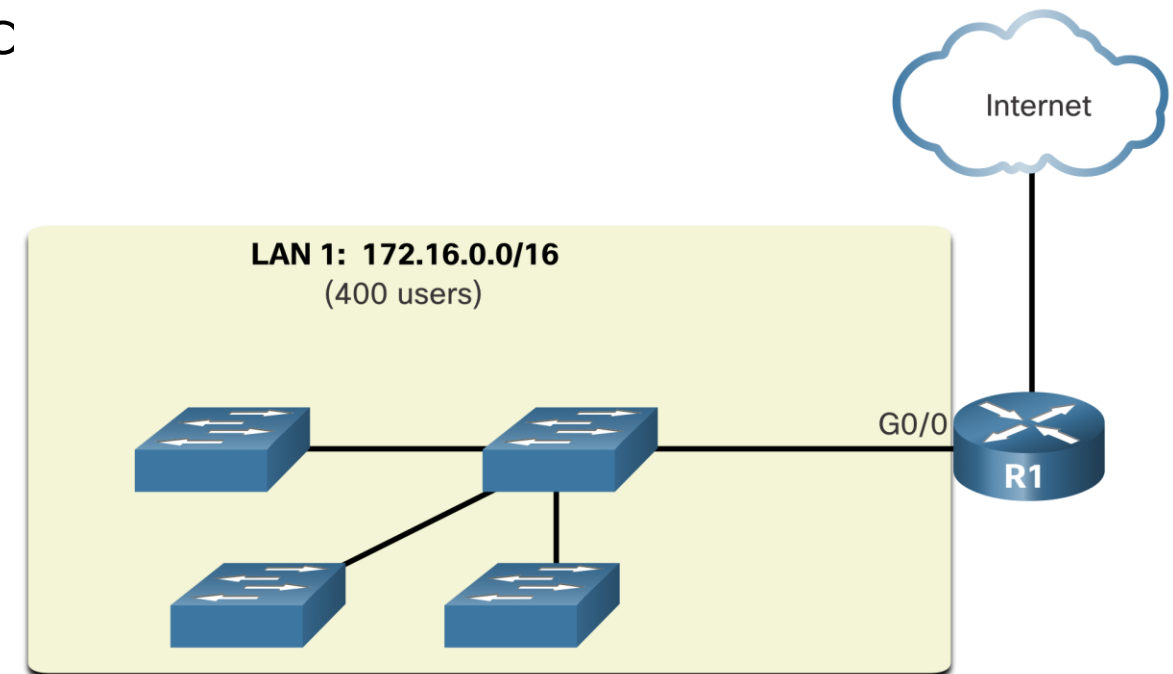
Broadcast domains

- A broadcast domain is a logical division of a computer network in which all hosts can reach each other by sending broadcast frames
- Switches forward broadcast frames through all their ports except the one from which the frames were received
- Routers, however, do not forward broadcast frames
- Therefore, any network connected to a router interface is called a broadcast domain



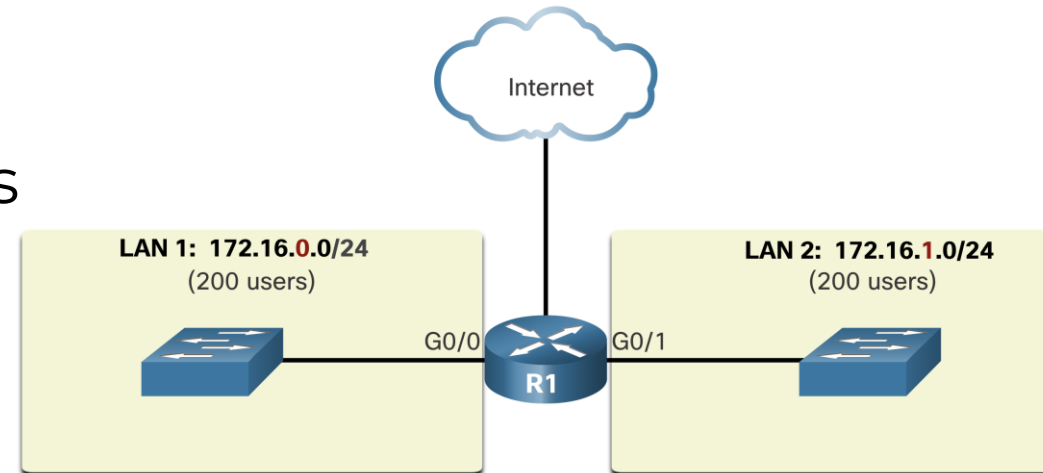
Problems with large broadcast domains

- A large broadcast domain is a network connecting many host
- These hosts can generate excessive number of broadcast frames (e.g., ARP) which can lead to
 - Slow network operations due to the amount of traffic generated
 - Slow device operations because each device needs to process each broadcast message



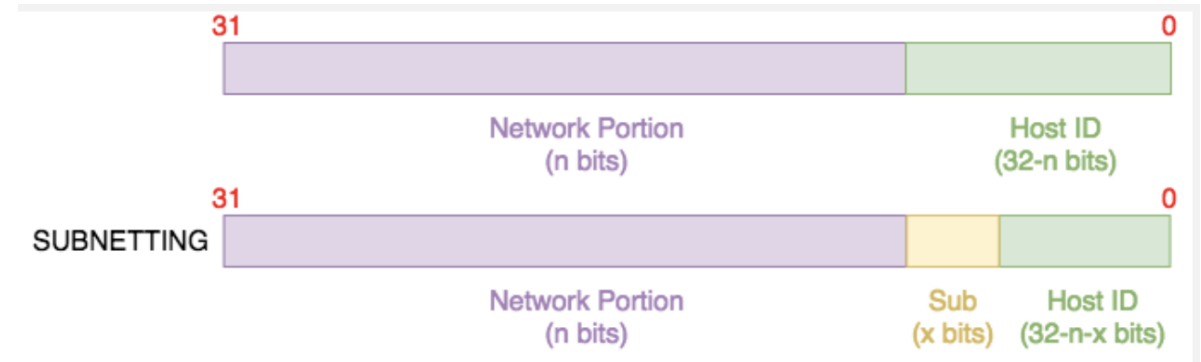
The solution - subnetting

- The solution is to reduce the size of the network to create smaller broadcast domains
- This is called subnetting
 - Dividing a network into smaller ones
- With subnetting, we can:
 - Reduce the overall network traffic
 - Implement security policies (e.g., subnets not/allowed to communicate together)
 - Reduce the number of affected devices due to hardware malfunctioning or security attack



Creating subnets

- We said that an IP address is made of **Netid** and **Hostid**
- To create subnets, we borrow bits from the **Hostid**
- The subnet mask is used to differentiate between the **Netid** portion and **Hostid** portion



Subnetting Class C networks

- **Hostid** has only 8 bits $\rightarrow 2^8 - 2 = 254$ host addresses

Bits borrowed from HostID	No. of Subnets	CIDR Notation	Subnet Mask
0	0	/24	255.255.255.0
1	2	/25	255.255.255.128
2	4	/26	255.255.255.192
3	8	/27	255.255.255.224
4	16	/28	255.255.255.240
5	32	/29	255.255.255.248
6	64	/30	255.255.255.252

- Note that /31 and /32 are unused as any network must have at least 2 hosts

Subnetting Class B networks

- **Hostid** has 16 bits
- $2^{16} - 2 = 65534$ host addresses

Bits borrowed from HostID	No. of Subnets	CIDR Notation	Subnet Mask
0	0	/16	255.255.0.0
1	2	/17	255.255.128.0
2	4	/18	255.255.192.0
3	8	/19	255.255.224.0
4	16	/20	255.255.240.0
5	32	/21	255.255.248.0
6	64	/22	255.255.252.0
7	128	/23	255.255.254.0
8	256	/24	255.255.255.0
...	...	...	...
14	16384	/30	255.255.255.252

Subnetting Class A networks

- **Hostid** has 24 bits
- $2^{24} - 2 = 16777216$ host addresses

Bits borrowed from HostID	No. of Subnets	CIDR Notation	Subnet Mask
0	0	/8	255.0.0.0
1	2	/9	255.128.0.0
2	4	/10	255.192.0.0
3	8	/11	255.224.0.0
4	16	/12	255.240.0.0
5	32	/13	255.248.0.0
6	64	/14	255.252.0.0
7	128	/15	255.254.0.0
8	256	/16	255.255.0.0
...	...	...	...
22	4194304	/30	255.255.255.252

Subnetting - example

- Given the network address 162.16.0.0/18, answer the following:
 1. How many subnets are there?
 2. What are the IP addresses of these subnets?
 3. What are the 1st and last host addresses of each subnet
 4. What is the broadcast address of each subnet

Subnetting - example

1. 162.16.0.0 is class B and the default subnet mask for Class B is /16
 - Since the new subnet mask is /18, it means that we borrowed 2 bits from the **Hostid**
 - Therefore, the number of subnets = $2^2 = 4$
2. Subnets' IP Addresses
 - 162.16.0.0 in binary $\rightarrow$ 10100010.00010000.00000000.00000000
 - 1st subnet: 10100010.00010000.00000000.00000000 $\rightarrow$ 162.16.0.0
 - 2nd subnet: 10100010. 00010000.01000000.00000000 $\rightarrow$ 162.16.64.0
 - 3rd subnet: 10100010. 00010000.10000000.00000000 $\rightarrow$ 162.16.128.0
 - 4th subnet: 10100010. 00010000.11000000.00000000 $\rightarrow$ 162.16.192.0

Subnetting - example

3. First and last host addresses in each subnet

Subnet	1 st Host Address	Last Host Address
162.16.0.0	162.16.0.1	162.16.63.254
162.16.64.0	162.16.64.1	162.16.127.254
162.16.128.0	162.16.128.1	162.16.191.254
162.16.192.0	162.16.192.1	162.16.255.254

4. Broadcast address of each subnet

Subnet	Broadcast Address
162.16.0.0	162.16.63.255
162.16.64.0	162.16.127.255
162.16.128.0	162.16.191.255
162.16.192.0	162.16.255.255

Subnetting - exercise

- Given the following network address 192.168.10.0/26, answer the following:
 1. How many subnets can we have?
 2. What are the addresses of these subnets?
 3. How many addressable hosts per subnet?
 4. What is the broadcast address on each subnet?

Subnetting - Exercise

1. 192.168.10.0/26 is class C and the default subnet mask for Class C is /24
 - Since the new subnet mask is /26, it means that we borrowed 2 bits from the **Hostid**
 - Therefore, the number of subnets = $2^2 = 4$

2. Subnets' IP Addresses

192.168.10.0 in binary → 110000000.10100010.00001010.00000000

1st subnet: 110000000.10100010.00001010.00000000 → 192.168.10.0

2nd subnet: 110000000.10100010.00001010.01000000 → 192.168.10.64

3rd subnet: 110000000.10100010.00001010.10000000 → 192.168.10.128

4th subnet: 110000000.10100010.00001010.11000000 → 192.168.10.192

Subnetting - Exercise

3. Since it is a class C address, we know that the **HostId** is 8 bits. Given that we borrowed 2 bits to create the subnets, we are left with only 6 bits for the **HostId**. Therefore, the number of addressable hosts per subnet is $2^6 - 2 = 62$
4. The broadcast address of each subnet can be obtained by setting all the **HostId** bits into 1

1st subnet: 11000000.10100010.00001010.00111111 → 192.168.10.63

2nd subnet: 11000000.10100010.00001010.01111111 → 192.168.10.127

3rd subnet: 11000000.10100010.00001010.10111111 → 192.168.10.191

4th subnet: 11000000.10100010.00001010.11111111 → 192.168.10.255

- Given the following IP address 201.102.5.11, answer the following:
 1. What is the IP address of the network to which this address belongs?
 2. How many bits we should borrow if we want to subnet this network into 4 subnets
 3. What are the addresses of these subnets?
 4. How many addressable hosts per subnet?
 5. What is the broadcast address on each subnet?

Subnetting - Exercise

1. 201.102.5.11 is class C and the default subnet mask for Class C is 255.255.255.0

We use the AND logic operator between the IP address and the subnet mask to get the Network IP address

201.102.5.11	→	1	1	0	0	1	0	0	1	.	1	1	0	0	1	1	0	0	.	0	0	0	0	0	1	0	1	.	0	0	0	0	1	0	1	1
255.255.255.0	→	1	1	1	1	1	1	1	1	.	1	1	1	1	1	1	1	1	.	1	1	1	1	1	1	1	1	.	0	0	0	0	0	0	0	
201.102.5.0	→	1	1	0	0	1	0	0	1	.	1	1	0	0	1	1	0	0	.	0	0	0	0	0	1	0	1	.	0	0	0	0	0	0	0	

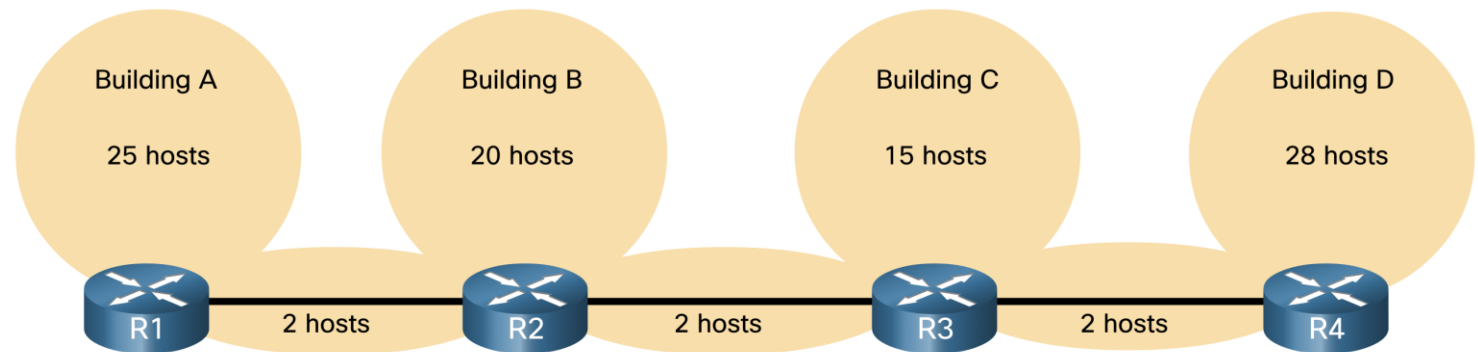
2. We should borrow 2 bits to create 4 subnets
3. 201.102.5.0, 201.102.5.64, 201.102.5.128, 201.102.5.192
4. $2^6 - 2 = 62$
5. 201.102.5.63, 201.102.5.127, 201.102.5.191, 201.102.5.255

Input A	Input B	A ^ B
0	0	0
0	1	0
1	0	0
1	1	1

Variable Length Subnet Mask (VLSM)

- Using traditional subnetting, the same subnet mask is applied to all subnets
 - This means that all subnets will have the same number of addresses

- What if we want subnets to have different number of addresses?



- This means that subnets will have different subnet masks
- This is known as VLSM – which is just subnetting a subnet

Variable Length Subnet Mask (VLSM)

- Using VLSM, we can create subnets with different sizes if their address ranges do not overlap
- e.g., assume we have the network 192.168.1.0/24 which we'll subnet into 4 subnets:

192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, 192.168.1.192/26

- Assume that we want to divide the subnet in blue into 2 smaller subnets. The addresses of the smaller subnets are:

192.168.1.128/27 and 192.168.1.160/27

- Now the network has 5 subnets with different sizes

192.168.1.0/26
192.168.1.64/26
192.168.1.192/26

192.168.1.128/27
192.168.1.160/27

THANK YOU